

Deep Learning for Parkinson's Disease: Identification, Classification and Lifestyle Recommendations

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Abstract- One such degenerative ailment is Parkinson's disease (PD) which is a brain disorder that progresses as an individual ages and affects many across the globe. Hence, early diagnosis and detection of the condition are necessary to help in the management of its symptoms and the speed at which it advances. This project proposes the development of a system for diagnosing PD and analysis of its symptoms via the imaging techniques and other patient data bereft of the biological examination. The system first employs a Convolutional Neural Network (CNN) algorithm to analyze certain brain images (MRI) for diagnosis of Parkinson's. Upon identification of the disease, the system also looks into the symptoms of the patient to grade the severity of the disease from early stages to more advanced cases. Moreover, it specifies peculiar types of parkinsonism, which can be characterized as a dominating tremor type, rigidity type, or mixed-parkinsonism. It also recommends treatment options and interventions, preventative measures, and other health professionals in order to help the health worker put together a total care package for the patient. This concept seeks to combine a typical imaging paradigm and symptom assessment so as to enhance the diagnosis and management of PD. Thus increasing the prognosis as well as the quality of life of patients.

Keywords- PD, CNN, Deep learning, Neurological disorders.

I. INTRODUCTION

PD refers to the neurodegenerative disease that causes motor and mental impairments to millions across the globe. An early diagnosis and accurate disease categorization is very important for effective treatment in which therapy can ameliorate the symptoms and enhance the patient's standard of living. However, most especially in the early stages, most of the traditional approaches for diagnosis revolve around clinical examination which may also be subjective and considerable symptoms are usually left out. This necessity has been the inspiration for the design of this project which aims at filling this gap. The main objective is to apply sophisticated designs in deep learning i.e. following the CNNs approaches in the Processing of Brain images and detection of PD in the initial stage, its severity and types as characterized by symptoms. The significance of this is covered by concrete disease management,

minimizing the reverse protective Mach bands that cause reactivity in patients with parkinsonism.

CNNs are deep neural networks that use convolutions or a specific linear operation as their mathematical foundation in one or more layers. Conceived in the vein of spatial organization, CNN uses a distinct method to avoid overfitting; it avoids creating fully connected networks where every neuron connects with every other neuron by enforcing limited connections among the neurons, somewhat akin to those occurring between proximities of neurons within the biological cortex, on the premise of highlighting regional features. CNNs were successfully applied in various JQE tasks, from image recognition and classification of faces and objects to disease detection and as such, are characterized by their presence of input, hidden, and output layers.

This system is based on brain image as well as symptoms analysis, which narrows down the scope for error occurred in diagnosis, enables management to take influenced actions earlier and assists taking care of patients with the levels those responsible for the care of patients consider useful. Furthermore, a recommendation system is integrated within the application for ease of access and treatment compliance. The impact of executing such a project can be a positive change on the clinical outcome of patients and elevate the standard of practice for patients.

II. RELATED WORKS

Numerous studies look into deep learning algorithms for PD identification. They look at current models and recommend enhancements like [1]. This article provides a thorough survey of the literature on deep learning-based techniques for PD prognosis, focusing on gait, motion, voice, and facial movement. It highlights several limitations, such as a lack of data and the models' interpretability, but also notes the notable distinctions from traditional methods.

[2] It was stated that the deep learning model created,

which uses the K-Nearest Neighbor technique and feature optimization based on evolutionary algorithms for grading Parkinsonism, achieved data accuracy in detection above 95%. With a loss of 0.12 and a 98% area under the curve and 0.90, the model performs more than 2% better than the current models. [3] This study examines 147 papers that discuss the use of artificial intelligence—specifically, Deep learning and machine learning in PD diagnosis, Alzheimer's disease, brain tumors, and epilepsy. In addition, it describes 22 frequently used databases and various feature extraction methods to determine the most accurate ways to diagnose brain diseases.

[4] In this work, the course of PD and medication management are examined using multi-modal data and reinforcement learning. It emphasizes how effective clinical treatment will be going forward. [5] With the use of a Bidirectional LSTM model, which increases detection accuracy in comparison to conventional models, this study explores voice analysis as a potential marker of PD and falls behind when examining speech characteristics such as articulations and frequency patterns.

[6] This study offers a sophisticated deep learning model that can identify early parkinsonism using premotor characteristics and sophisticated imaging, with an accuracy level of up to 96.45%. However, it outperforms twelve machine learning algorithms that use boosting to examine the significance of characteristics. Using foot sensor-based walking data, this study suggests a Convolutional Neural Network (CNN) for PD identification and progression evaluation. After hyper-parameter adjustment, the method outperforms previous studies on the known dataset and accurately classifies impacted individuals.

[7] This study uses Random Forest classifiers and deep neural nets using UCI's Telemonitoring Vocal Data Set to predict the severity of a particular PD case, showing tremendous promise for early intervention. This study uses Random Forest classifiers and deep neural networks to predict the severity of Parkinsonism with a high potential for prompt intervention, using the Telemonitoring Vowel Dataset from UCI. [8] This study uses a CNN on T2-weighted magnetic resonance imaging (MRI) of the brain to detect PD. When compared to state-of-the-art methods, mid-brain slices from the PPMI dataset have been demonstrated to increase accuracy, sensitivity, specificity, and AUC by 3% to 9%.

[9] DenseNet is used to classify the images, and an ensemble of BOA with LBG is proposed for effective compression. The settings in the aforementioned process will be adjusted using the Adam optimizer. According to simulation results, CMOS sensor image processing performed better than more modern ones. [10] In order to

achieve high accuracy using improved algorithms, this work uses transfer learning with VGG16 and ResNet101 to categorize neurodegenerative disorders such as Parkinson's, Alzheimer's, and cerebral ataxia. When used with Flask, the model offers a productive and easy-to-use tool for precise disease classification and better patient care.

[11] This work presents three speech analysis-based algorithms for early PD identification, outperforming previous approaches on the pc-Gita dataset. [12] This article outlines a computer-aided diagnosis technique that can detect PD from SPECT images with a 98.47% accuracy rate using bilinear pooling and pre-trained convolutional neural networks. The system's enhanced diagnostic accuracy and greater objectivity are very beneficial to doctors.

[13] The article presents the DMVDA-based system, which classifies vocal data highly related to PD symptoms using acoustic-based deep learning algorithms.

III. PROPOSED METHODOLOGY

1. Problem Definition:

The purpose is to stop PD progression, early detection and precise categorization are essential. Conventional diagnostic techniques, which depend on motor evaluations and clinical assessments, are time-consuming and biased. A quicker, self-sufficient system is required to reduce the strain on patients and healthcare personnel. This research aims to develop a deep learning system that can reliably identify and stage Parkinson's disease by analyzing biomarkers and clinical data. The program will help doctors diagnose and treat Parkinson's disease by classifying it and distinguishing it from other brain-related disorders using artificial intelligence (AI), more especially machine learning techniques..

2. Dataset:

The dataset was sourced from Kaggle and consists of two categories of brain MRIs: patients diagnosed with PD versus healthy individuals. Each MRI scan has been labeled with the diagnostic status of the subject, thus allowing a binary classification (Parkinson's versus normal). The set contains high-resolution MRI scans that highlight the pathological features on the brain that will allow the discrimination between Parkinson's and non-Parkinson's diseases. All images underwent preprocessing, which included resizing and normalization, to guarantee the inputs to training and test are of equal dimensions. It is also categorized with its images and labels in folders to ease the data load and split for creation purposes. This data set is, therefore, an excellently curated dataset for training the CNN algorithm for building accurate automated diagnostic aids for PD.

Brain MRI pictures from two groups—healthy people and Parkinson's patients—are included in the Kaggle dataset. They are labeled for binary classification (normal vs. Parkinson's). In order to differentiate PD, high-resolution scans show anomalies. Scaling, normalization, and uniform resizing are preprocessing techniques that ensure a constant input size. For ease of loading, data is arranged in labeled folders. This configuration is perfect for CNN training.

The collection provides a variety of brain MRI scans that show important pathological characteristics, making it possible to effectively learn about illness patterns. For CNN-based automated diagnostic models, its systematic labelling and preprocessing guarantee a smooth integration procedure.

3. System Architecture

In Figure 1, Architecture diagram expresses a process of classifying Parkinson disease using brain MRI images. The process starts with a brain MRI image dataset containing image labels that will be uploaded and images that will be extracted. After preprocessing, the data is divided into test and training sets, which can then be further enhanced. A deep learning network is trained, and the output produced by this network aims to lead to Parkinson classification. Based upon results, it identifies normal cases versus various forms of Parkinsonism (i.e., vascular, drug-induced, and idiopathic) along with suggestions on accuracy and classification results.

3.3.1 Module Description

A. Image Acquisition:

Beginning with the collection of MRIs for PD patients,

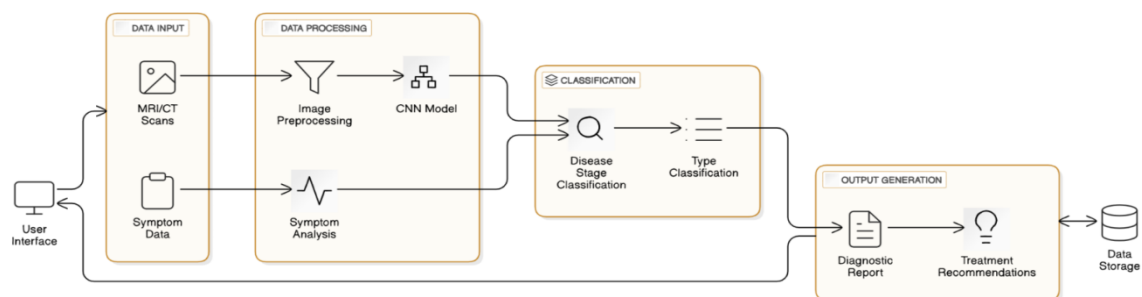


Figure 1: Architecture Diagram

d. MRI Classification:

We are performing classification operations utilizing deep learning algorithms to detect Parkinson's illness. Determining whether PD is aberrant or normal is believed to depend heavily on how the planned study classifies the disorder. The extracted and selected features will be used in the classification stage once CNN has been applied to the final feature vector. Classification will be used in the training and testing stages of the CNN structure.

the suggested methods have been put into practice. This methodology's design is based on the extraction of images of PD with indicators of good accuracy. It includes several stages, including image acquisition, edge identification in the images, and classification of the condition.

B. Image Pre-Processing:

Basic image preprocessing is the main emphasis of this module in order to produce acceptable and comprehensible processing inputs. To improve image quality by lowering noise, operations such as grayscale conversion, filtering, sharpening, smoothing, edging, and segmentation are used. Only shades of gray are used in grayscale photographs, and brightness is represented by pixel intensity. While smoothing lowers noise for cleaner results, filtering improves edges, and sharpening increases contrast and detail visibility. These methods efficiently get pictures ready for additional processing.

C. Feature Extraction:

In order to classify PD, this module analyzes segmented images and extracts important brain imaging characteristics. Robust characteristics that are essential for precise computer vision and machine learning categorization are found through feature extraction. By concentrating on the most active aspects, it seeks to optimize performance. It also streamlines data for effective analysis by reducing dimensionality.

E. Convolutional Neural Network:

Deep neural networks that process data using convolution, a kind of linear operation, are known as convolutional neural networks (CNNs). CNNs, which are composed of input, output, and numerous hidden layers, are excellent at tasks like face recognition, object classification, and disease diagnosis. They transmit outputs between layers and extract features by combining pooling and convolution processes. Multilayer perceptrons, in which neurons are interconnected in fully connected networks,

are regularized variants of CNNs. The identification and classification of images is a common application for these networks.

4. Performance Metrics

Performance metrics are quantitative assessments of the model's efficacy, correctness, and generalizability.

1. Accuracy

Accuracy is defined as the ratio of accurate predictions to total predictions, and is one of the most straightforward metrics to employ for classification.

2. Precision

Precision is the solution to the limitation of accuracy. Using precision, you may calculate the proportion of accurate forecasts that materialized. The calculation involves dividing the total number of false positives (erroneous positives) by the number of true positives (correct forecasts).

3. Recall Or Sensitivity

Though it assumes to compute the unknown percentage of true positives, it is comparable to Precision. True positives or false negatives are predictions that are accurate relative to the total number of positives, either accurately projected as positive or wrongly projected as negative.

4. F scores

To evaluate binary classification models, the F Score (sometimes called the F1 Score) is calculated based on the predictions made for the positive class. Its calculation takes advantage of recall and precision. Both recall and accuracy are accurately reflected in a single score.

IV. RESULT AND DISCUSSION

4.1 Experimental Setup

The entire project was built on Windows 10 environment in a system with Processor Intel ,Speed-1.1 GHz configuration, 8GB RAM and 512GB SSD. The application is based on python 3.10 and built using jupyter notebook from anaconda version 3. The sklearn package for model creation and selection for the prediction system is used. Flask provides a framework for building the web application through APIs and HTML for building the web application.

4.2 Observation

Model Architecture

To specifically diagnose PD, the CNN model was created. The model receives 200x200 pixel RGB images. It uses a sequence of convolutional layers and MaxPooling layers to assist extract information from the pictures. The last layer uses softmax activation to create probabilities for each class (Normal and Parkinson's) after the feature maps have been flattened and input into dense layers.

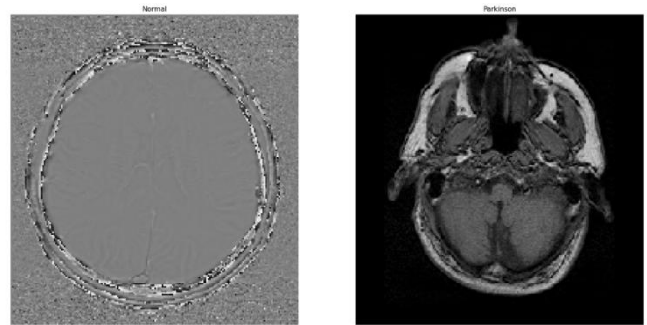


Figure 3: MRI Scans of Human Brain

Training and Evaluation

The model generates data generators for training, validation, and testing stages using ImageDataGenerator. To enhance learning and classification, it is trained using the RMSprop optimizer and categorical cross-entropy loss function. Performance measures are used in the model's evaluation.

Performance Analysis

The loss and accuracy plots illustrate that the model has learnt well. The curves show decreasing loss over epochs, indicating the model is learning with more updates. Similarly, the accuracy increased in each epoch, which means the model is converging in the right direction. However, we see slight fluctuations in the curves, which occur primarily due to the random nature of our optimization process, i.e. the initialization of our random weights. It can also happen due to the complexity of the data, which makes it harder for the model to learn perfectly.

Loss Curve

The model is learning and getting better, as evidenced by the loss curve's declining tendency over epochs.

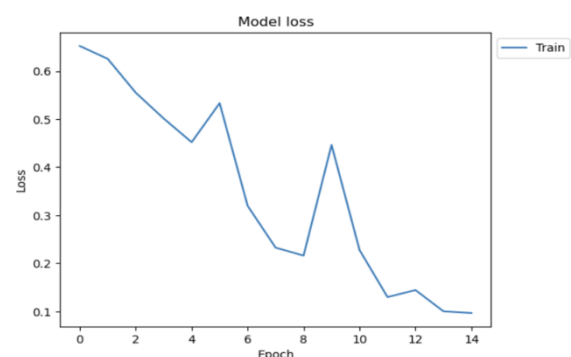


Figure 4: Model Loss

Accuracy Curve

The model is performing better, as seen by the accuracy curve's growing trend over epochs.

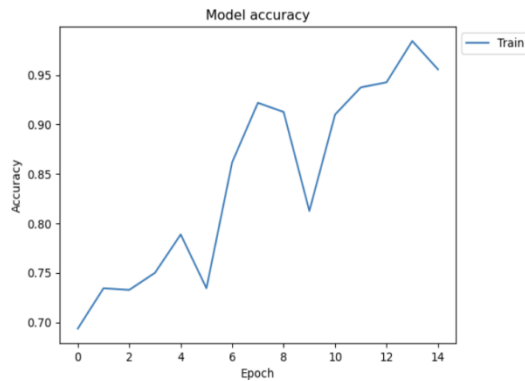


Figure 5: Model Accuracy

V.CONCLUSION AND FUTURE WORK

The suggested approach, which makes use of CNN superiority and effectiveness as image processing tools, marks a significant advancement in the early diagnosis and detection of PD. The automation of the procedure enables the integration of brain imaging and symptom evaluations to enhance the effectiveness and precision of diagnosis, paving the way for customized treatment regimens for every patient. In addition to addressing some of the obvious shortcomings of the prevalent traditional diagnostic approaches, a holistic approach would surely provide a healthcare practitioner with a potent instrument to decrease subjectivity and expedite diagnostic turnaround times. In addition to stimulating reflective service starting techniques and offering significant insights for further research, the system will be of tremendous use to patients.

Future developments in Parkinson's disease prediction systems may take into account hereditary variables and sophisticated gait analysis to enhance forecasts, including the disease's progression to dementia. Wearable technology-based real-time brain assessments could improve disease tracking by continuously monitoring changes in the internal and motor systems. The system's usefulness would be increased by extending deep learning models to take into account various age groups and surroundings. By including treatment planning into the application, therapy recommendations based on Parkinson's subtype and severity could enhance diagnosis and patient care. Clarifying predictions using Explainable AI (XAI) would boost physician confidence and

acceptance of AI-assisted diagnosis. This improved method would be extremely useful in clinical settings since it would facilitate patient monitoring, early detection, and tailored medication.

REFERENCES

- [1]. Pagano, G., Monnet, A., Reyes, A. et al. Sustained effect of prasinezumab on Parkinson's disease motor progression in the open-label extension of the PASADENA trial. *Nat Med* (2024)
- [2]. S. M. Abdullah et al., "Deep Transfer Learning Based Parkinson's Disease Detection Using Optimized Feature Selection," in *IEEE Access*, vol. 11, pp. 3511-3524, 2023, doi: 10.1109/ACCESS.2023.3233969.
- [3]. P. Khan et al., "Machine Learning and Deep Learning Approaches for Brain Disease Diagnosis: Principles and Recent Advances," in *IEEE Access*, vol. 9, pp. 37622-37655, 2021, doi: 10.1109/ACCESS.2021.3062484.
- [4]. Beucler, Nathan. (2024). Parkinson's disease. *The New-England Medical Review and Journal*. 10.1056/NEJMc2410843.
- [5]. W. Wang, J. Lee, F. Harrou and Y. Sun, "Early Detection of Parkinson's Disease Using Deep Learning and Machine Learning," in *IEEE Access*, vol. 8, pp. 147635-147646, 2020, doi: 10.1109/ACCESS.2020.3016062.
- [6]. Dogan, Y. An innovative approach for parkinson's disease diagnosis using CNN, NCA, and SVM. *Neural Comput&Applic* 36, 20089–20110 (2024).
- [7]. H. Gunduz, "Deep Learning-Based Parkinson's Disease Classification Using Vocal Feature Sets," in *IEEE Access*, vol. 7, pp. 115540-115551, 2019, doi: 10.1109/ACCESS.2019.2936564..
- [8]. Pratihari, Ruchira, and Ravi Sankar. 2024. "Advancements in Parkinson's Disease Diagnosis: A Comprehensive Survey on Biomarker Integration and Machine Learning" *Computers* 13, no. 11: 293.
- [9]. Palani, Vinayagam, et al. "Deep Learning Based Compression with Classification Model on CMOS Image Sensors." *Traitement du Signal* 40.3 (2023).
- [10]. R. K. Mahendran, R. Aishwarya, R. Abinayapriya and P. Kumar, "Deep Transfer Learning Based Diagnosis of Multiple Neurodegenerative Disorders," 2024 International Conference on Emerging Smart Computing and Informatics (ESCI), Pune, India, 2024, pp. 1-4, doi: 10.1109/ESCI59607.2024.10497320.
- [11]. Ianculescu, Marilena, Corina Petean, Virginia Sandulescu, Adriana Alexandru, and Ana-Mihaela Vasilevschi. 2024. "Early Detection of Parkinson's Disease Using AI Techniques and Image Analysis" *Diagnostics* 14, no. 23: 2615.
- [12]. Chowdhary, Chiranjil Lal, and R. Srivatsan. "Non-invasive Detection of Parkinson's Disease Using Deep Learning." *International Journal of Image, Graphics and Signal Processing* 14.2 (2022): 38.
- [13]. Nagasubramanian, G., Sankayya, M. Multi-Variate vocal data analysis for Detection of Parkinson disease using Deep Learning. *Neural Comput&Applic* 33, 4849–4864 (2021).